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MSSV: 089206010393

1. Nạp dự liệu iris vào bộ nhớ và hiển thị thông tin về số thuộc tính (dimensionality), số objects (size).

In [1]:
```
from sklearn.datasets import load_wine
```

In [2]:
```
wine = load_wine(as_frame=True)
```

In [3]:
```
type(wine)
```

Out[3]:
```
sklearn.utils._bunch.Bunch
```

In [4]:
```
print(wine)
```

```
{'data':      alcohol  malic_acid   ash  alcalinity_of_ash  magnesium  total
_phenols  \
0      14.23        1.71  2.43               15.6      127.0           2.80
1      13.20        1.78  2.14               11.2      100.0           2.65
2      13.16        2.36  2.67               18.6      101.0           2.80
3      14.37        1.95  2.50               16.8      113.0           3.85
4      13.24        2.59  2.87               21.0      118.0           2.80
..       ...         ...   ...                ...        ...            ...
173    13.71        5.65  2.45               20.5       95.0           1.68
174    13.40        3.91  2.48               23.0      102.0           1.80
175    13.27        4.28  2.26               20.0      120.0           1.59
176    13.17        2.59  2.37               20.0      120.0           1.65
177    14.13        4.10  2.74               24.5       96.0           2.05

     flavanoids  nonflavanoid_phenols  proanthocyanins  color_intensity   hu
e  \
0          3.06                  0.28             2.29             5.64  1.0
4
1          2.76                  0.26             1.28             4.38  1.0
5
2          3.24                  0.30             2.81             5.68  1.0
3
3          3.49                  0.24             2.18             7.80  0.8
6
4          2.69                  0.39             1.82             4.32  1.0
4
..          ...                   ...              ...              ...
...
173        0.61                  0.52             1.06             7.70  0.6
4
174        0.75                  0.43             1.41             7.30  0.7
0
175        0.69                  0.43             1.35            10.20  0.5
9
176        0.68                  0.53             1.46             9.30  0.6
0
177        0.76                  0.56             1.35             9.20  0.6
1

     od280/od315_of_diluted_wines  proline
0                            3.92   1065.0
1                            3.40   1050.0
2                            3.17   1185.0
3                            3.45   1480.0
4                            2.93    735.0
..                            ...      ...
173                          1.74    740.0
174                          1.56    750.0
175                          1.56    835.0
176                          1.62    840.0
177                          1.60    560.0

[178 rows x 13 columns], 'target': 0      0
1      0
2      0
3      0
```

```
4      0
      ..
173    2
174    2
175    2
176    2
177    2
Name: target, Length: 178, dtype: int64, 'frame':      alcohol  malic_acid
ash  alcalinity_of_ash  magnesium  total_phenols  \
0      14.23        1.71  2.43               15.6      127.0           2.80
1      13.20        1.78  2.14               11.2      100.0           2.65
2      13.16        2.36  2.67               18.6      101.0           2.80
3      14.37        1.95  2.50               16.8      113.0           3.85
4      13.24        2.59  2.87               21.0      118.0           2.80
..       ...         ...   ...                ...        ...           ...
173    13.71        5.65  2.45               20.5       95.0           1.68
174    13.40        3.91  2.48               23.0      102.0           1.80
175    13.27        4.28  2.26               20.0      120.0           1.59
176    13.17        2.59  2.37               20.0      120.0           1.65
177    14.13        4.10  2.74               24.5       96.0           2.05

     flavanoids  nonflavanoid_phenols  proanthocyanins  color_intensity   hu
e  \
0          3.06                  0.28             2.29             5.64  1.0
4
1          2.76                  0.26             1.28             4.38  1.0
5
2          3.24                  0.30             2.81             5.68  1.0
3
3          3.49                  0.24             2.18             7.80  0.8
6
4          2.69                  0.39             1.82             4.32  1.0
4
..          ...                   ...              ...              ...
...
173        0.61                  0.52             1.06             7.70  0.6
4
174        0.75                  0.43             1.41             7.30  0.7
0
175        0.69                  0.43             1.35            10.20  0.5
9
176        0.68                  0.53             1.46             9.30  0.6
0
177        0.76                  0.56             1.35             9.20  0.6
1

     od280/od315_of_diluted_wines  proline  target
0                            3.92   1065.0       0
1                            3.40   1050.0       0
2                            3.17   1185.0       0
3                            3.45   1480.0       0
4                            2.93    735.0       0
..                            ...      ...     ...
173                          1.74    740.0       2
174                          1.56    750.0       2
175                          1.56    835.0       2
```

```
176                          1.62    840.0       2
177                          1.60    560.0       2

[178 rows x 14 columns], 'target_names': array(['class_0', 'class_1', 'class
_2'], dtype='<U7'), 'DESCR': '.. _wine_dataset:\n\nWine recognition dataset
\n------------------------\n\n**Data Set Characteristics:**\n\n:Number of In
stances: 178\n:Number of Attributes: 13 numeric, predictive attributes and t
he class\n:Attribute Information:\n    - Alcohol\n    - Malic acid\n    - As
h\n    - Alcalinity of ash\n    - Magnesium\n    - Total phenols\n    - Flav
anoids\n    - Nonflavanoid phenols\n    - Proanthocyanins\n    - Color inten
sity\n    - Hue\n    - OD280/OD315 of diluted wines\n    - Proline\n    - cl
ass:\n        - class_0\n        - class_1\n        - class_2\n\n:Summary St
atistics:\n\n============================= ==== ===== ======= =====\n
Min   Max   Mean     SD\n============================= ==== ===== ======= ==
===\nAlcohol:                     11.0  14.8    13.0   0.8\nMalic Acid:
0.74  5.80    2.34  1.12\nAsh:                           1.36  3.23    2.36
0.27\nAlcalinity of Ash:            10.6  30.0    19.5   3.3\nMagnesium:
70.0 162.0    99.7  14.3\nTotal Phenols:                0.98  3.88    2.29
0.63\nFlavanoids:                   0.34  5.08    2.03  1.00\nNonflavanoid P
henols:         0.13  0.66    0.36  0.12\nProanthocyanins:              0.41
3.58    1.59  0.57\nColour Intensity:              1.3  13.0     5.1   2.3\n
Hue:                           0.48  1.71    0.96  0.23\nOD280/OD315 of dilut
ed wines: 1.27  4.00    2.61  0.71\nProline:                       278  1680
746   315\n============================= ==== ===== ======= =====\n\n:Missin
g Attribute Values: None\n:Class Distribution: class_0 (59), class_1 (71), c
lass_2 (48)\n:Creator: R.A. Fisher\n:Donor: Michael Marshall (MARSHALL%PLU@i
o.arc.nasa.gov)\n:Date: July, 1988\n\nThis is a copy of UCI ML Wine recognit
ion datasets.\nhttps://archive.ics.uci.edu/ml/machine-learning-databases/win
e/wine.data\n\nThe data is the results of a chemical analysis of wines grown
in the same\nregion in Italy by three different cultivators. There are thirt
een different\nmeasurements taken for different constituents found in the th
ree types of\nwine.\n\nOriginal Owners:\n\nForina, M. et al, PARVUS -\nAn Ex
tendible Package for Data Exploration, Classification and Correlation.\nInst
itute of Pharmaceutical and Food Analysis and Technologies,\nVia Brigata Sal
erno, 16147 Genoa, Italy.\n\nCitation:\n\nLichman, M. (2013). UCI Machine Le
arning Repository\n[https://archive.ics.uci.edu/ml]. Irvine, CA: University
of California,\nSchool of Information and Computer Science.\n\n.. dropdown::
References\n\n    (1) S. Aeberhard, D. Coomans and O. de Vel,\n    Compariso
n of Classifiers in High Dimensional Settings,\n    Tech. Rep. no. 92-02, (1
992), Dept. of Computer Science and Dept. of\n    Mathematics and Statistic
s, James Cook University of North Queensland.\n    (Also submitted to Techno
metrics).\n\n    The data was used with many others for comparing various\n
classifiers. The classes are separable, though only RDA\n    has achieved 10
0% correct classification.\n    (RDA : 100%, QDA 99.4%, LDA 98.9%, 1NN 96.1%
(z-transformed data))\n    (All results using the leave-one-out technique)\n
\n    (2) S. Aeberhard, D. Coomans and O. de Vel,\n    "THE CLASSIFICATION P
ERFORMANCE OF RDA"\n    Tech. Rep. no. 92-01, (1992), Dept. of Computer Scie
nce and Dept. of\n    Mathematics and Statistics, James Cook University of N
orth Queensland.\n    (Also submitted to Journal of Chemometrics).\n', 'feat
ure_names': ['alcohol', 'malic_acid', 'ash', 'alcalinity_of_ash', 'magnesiu
m', 'total_phenols', 'flavanoids', 'nonflavanoid_phenols', 'proanthocyanin
s', 'color_intensity', 'hue', 'od280/od315_of_diluted_wines', 'proline']}
```

In [5]:
```
wine.data
```

Out[5]:

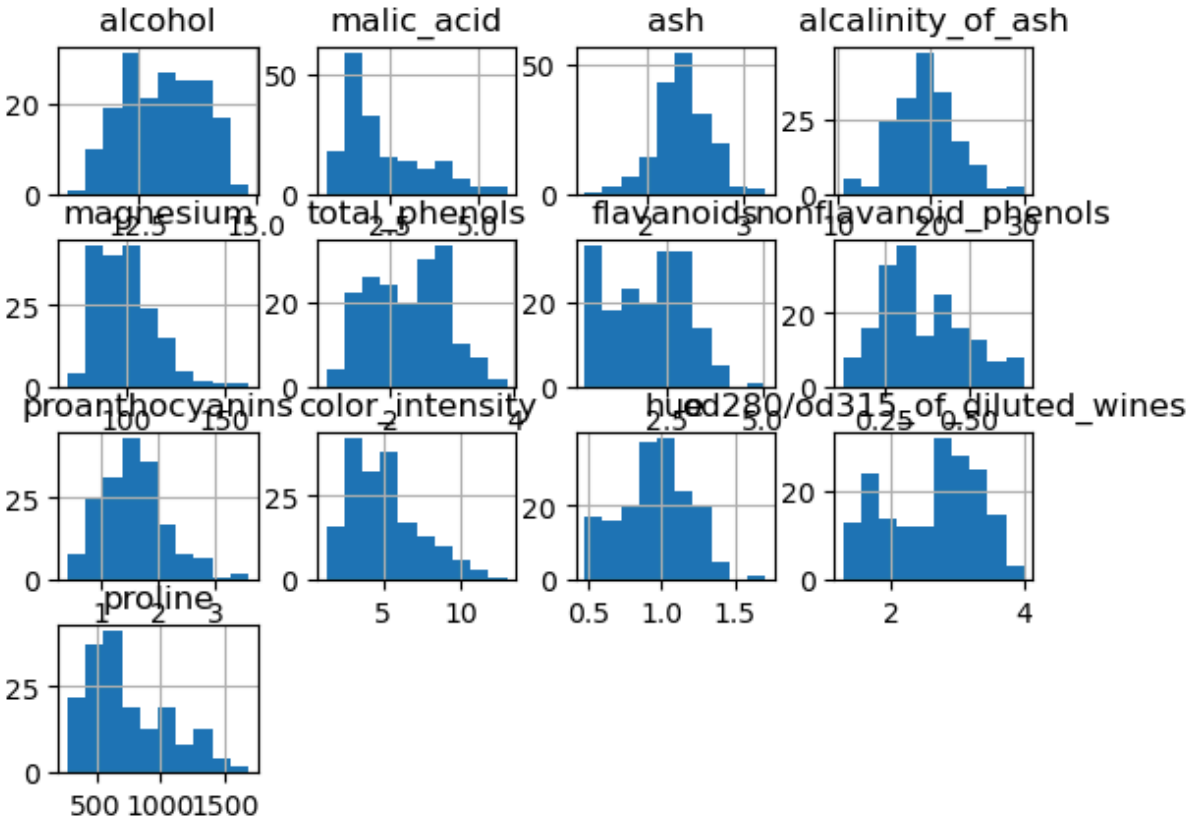
	alcohol	malic_acid	ash	alcalinity_of_ash	magnesium	total_phenols	flavanoids
0	14.23	1.71	2.43	15.6	127.0	2.80	3.06
1	13.20	1.78	2.14	11.2	100.0	2.65	2.76
2	13.16	2.36	2.67	18.6	101.0	2.80	3.24
3	14.37	1.95	2.50	16.8	113.0	3.85	3.49
4	13.24	2.59	2.87	21.0	118.0	2.80	2.69
...	...	...	...	...	...	...	...
173	13.71	5.65	2.45	20.5	95.0	1.68	0.61
174	13.40	3.91	2.48	23.0	102.0	1.80	0.75
175	13.27	4.28	2.26	20.0	120.0	1.59	0.69
176	13.17	2.59	2.37	20.0	120.0	1.65	0.68
177	14.13	4.10	2.74	24.5	96.0	2.05	0.76

178 rows × 13 columns

In [6]:
```
from sklearn.preprocessing import StandardScaler
```

In [7]:
```
wine.data.hist()
```

Out[7]:
```
array([[<Axes: title={'center': 'alcohol'}>,
        <Axes: title={'center': 'malic_acid'}>,
        <Axes: title={'center': 'ash'}>,
        <Axes: title={'center': 'alcalinity_of_ash'}>],
       [<Axes: title={'center': 'magnesium'}>,
        <Axes: title={'center': 'total_phenols'}>,
        <Axes: title={'center': 'flavanoids'}>,
        <Axes: title={'center': 'nonflavanoid_phenols'}>],
       [<Axes: title={'center': 'proanthocyanins'}>,
        <Axes: title={'center': 'color_intensity'}>,
        <Axes: title={'center': 'hue'}>,
        <Axes: title={'center': 'od280/od315_of_diluted_wines'}>],
       [<Axes: title={'center': 'proline'}>, <Axes: >, <Axes: >,
        <Axes: >]], dtype=object)
```



In [8]:
```
wine_data = wine.data
```

In [9]:
```
wine_target = wine.target
```

In [10]:
```
wine_target.value_counts()
```

Out[10]:
```
target
1    71
0    59
2    48
Name: count, dtype: int64
```

3. Tách dữ liệu thành hai phần: training data(70%) và test data (30%

In [11]:
```
from sklearn.model_selection import train_test_split
train_x, test_x, train_y, test_y = train_test_split(wine_data, wine_target,
```

In [12]:
```
train_x
```

Out[12]:

	alcohol	malic_acid	ash	alcalinity_of_ash	magnesium	total_phenols	flavanoids
0	14.23	1.71	2.43	15.6	127.0	2.80	3.06
35	13.48	1.81	2.41	20.5	100.0	2.70	2.98
161	13.69	3.26	2.54	20.0	107.0	1.83	0.56
62	13.67	1.25	1.92	18.0	94.0	2.10	1.79
31	13.58	1.66	2.36	19.1	106.0	2.86	3.19
...	...	...	...	...	...	...	...
101	12.60	1.34	1.90	18.5	88.0	1.45	1.36
26	13.39	1.77	2.62	16.1	93.0	2.85	2.94
11	14.12	1.48	2.32	16.8	95.0	2.20	2.43
71	13.86	1.51	2.67	25.0	86.0	2.95	2.86
37	13.05	1.65	2.55	18.0	98.0	2.45	2.43

124 rows × 13 columns

In [13]:
```
len(train_y)
```

Out[13]: 124

4. Chuẩn hóa dữ liệu với StandardScaler

In [14]:
```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
fitted_train_x = scaler.fit_transform(train_x)
fitted_test_x = scaler.fit_transform(test_x)
```

5. Tạo và huấn luyện mô hình với training data

In [15]:
```
from sklearn.tree import DecisionTreeClassifier
tree_clf = DecisionTreeClassifier(max_depth=2)
```

In [16]:
```
tree_clf.fit(fitted_train_x, train_y)
```

Out[16]:

▾ DecisionTreeClassifier

▶ Parameters

In [17]:
```
fitted_test_x = scaler.fit_transform(test_x)
```

In [18]:
```
pred = tree_clf.predict(fitted_test_x)
```

In [19]:
```
from sklearn.metrics import accuracy_score, precision_score, recall_score, c
```

In [24]:
```
accuracy_score(test_y, pred)
```

Out[24]:
```
0.7592592592592593
```

In [25]:
```
confusion_matrix(test_y, pred)
```

Out[25]:
```
array([[14,  1,  0],
       [ 2, 20,  6],
       [ 4,  0,  7]])
```

In [26]:
```
print(classification_report(test_y, pred))
```

```
              precision    recall  f1-score   support

           0       0.70      0.93      0.80        15
           1       0.95      0.71      0.82        28
           2       0.54      0.64      0.58        11

    accuracy                           0.76        54
   macro avg       0.73      0.76      0.73        54
weighted avg       0.80      0.76      0.76        54
```

In [27]:
```
from sklearn.metrics import ConfusionMatrixDisplay
import numpy as np
import matplotlib.pyplot as plt
```

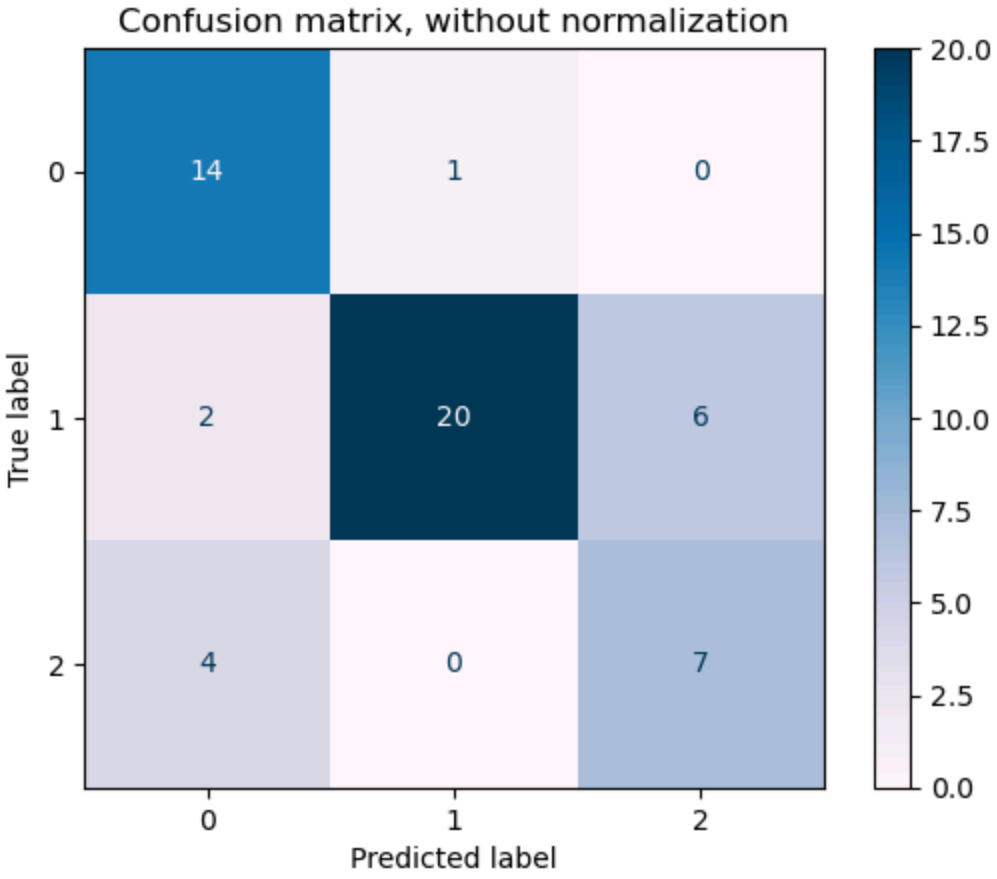
In [29]:
```
np.set_printoptions(precision=2)

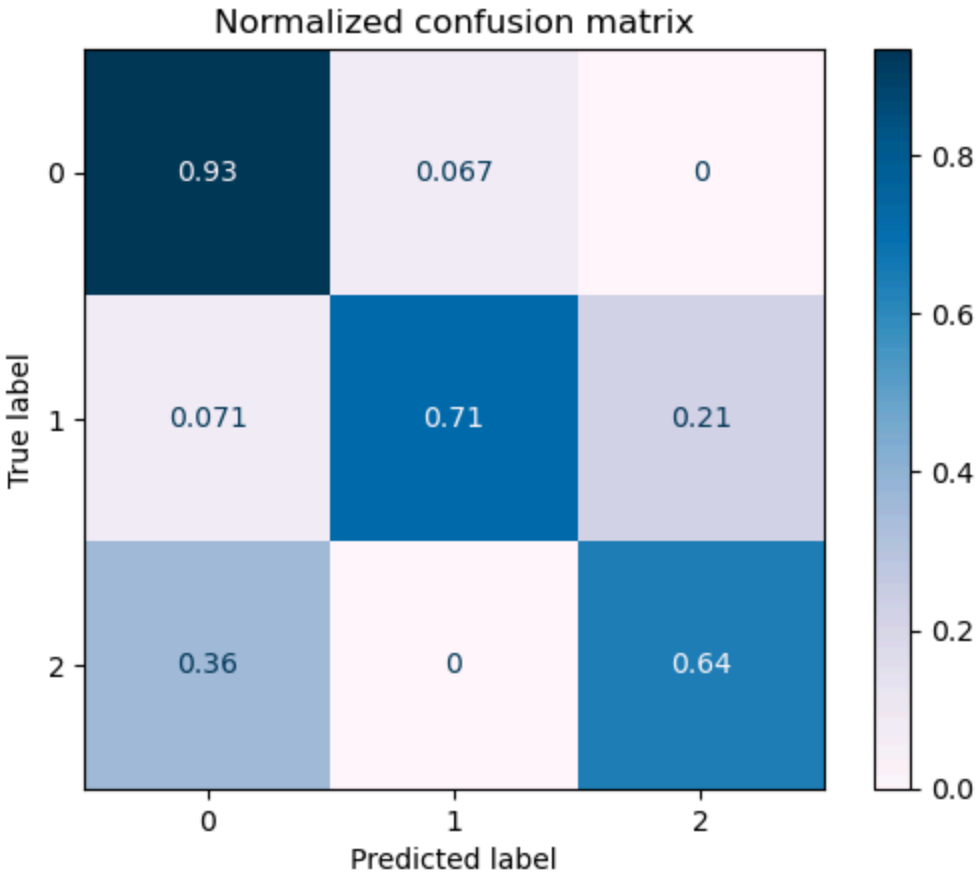
titles_options = [
    ("Confusion matrix, without normalization", None),
    ("Normalized confusion matrix", "true"),
]
for title, normalize in titles_options:
    disp = ConfusionMatrixDisplay.from_estimator(
        tree_clf,
        fitted_test_x,
        test_y,
        cmap=plt.cm.PuBu,
        normalize=normalize,
    )
    disp.ax_.set_title(title)

    print(title)
    print(disp.confusion_matrix)

plt.show()
```

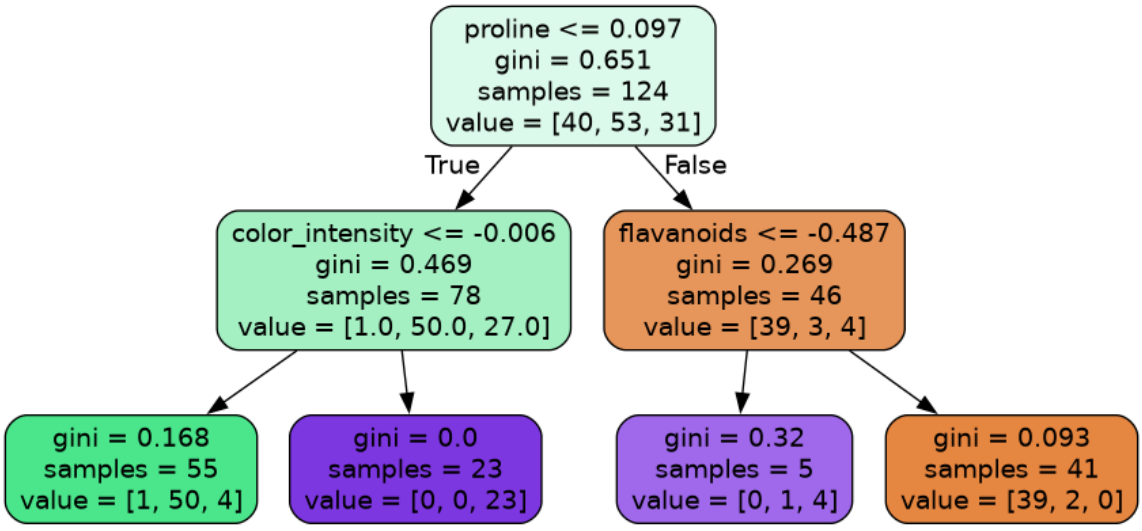
```
Confusion matrix, without normalization
[[14  1  0]
 [ 2 20  6]
 [ 4  0  7]]
Normalized confusion matrix
[[0.93 0.07 0.  ]
 [0.07 0.71 0.21]
 [0.36 0.   0.64]]
```





In [ ]:
```
from sklearn.tree import export_graphviz
export_graphviz(
    tree_clf,
    out_file="wine.dot",
    feature_names=train_x.columns,
    rounded=True,
    filled=True
)
```

In [ ]:
```
!dot -Tpng wine.dot -o wine.png
```



In [ ]: